directly exert any force: only pressure differences, like what your ears feel when you're underwater, result in a pressure force. Instead, the force exerted by the cosmological constant is purely a gravitational force.

8. Familiar magnets always have both a north and a south pole. By contrast, grand unified theories suggest that there may be particles that are like a purely north or purely south magnetic pole. Such particles are called monopoles and they could have a major impact on standard big bang cosmology. They have never been observed.

9. Guth and Tye recognized that a supercooled Higgs field would act like a cosmological constant, a realization that had been made earlier by Martinus Veltman and others. In fact, Tye has told me that were it not for a page limit in *Physical Review Letters,* the journal to which he and Guth submitted their paper, they would not have struck a final sentence noting that their model would entail a period of exponential expansion. But Tye also notes that it was Guth's achievement to realize the important cosmological implications of a period of exponential expansion (to be discussed later in this and in the next chapter), and thereby put inflation front and center on cosmologists' maps.

In the sometimes convoluted history of discovery, the Russian physicist Alexei Starobinsky had, a few years earlier, found a different means of generating what we now call inflationary expansion, work described in a paper that was not widely known among western scientists. However, Starobinsky did not emphasize that a period of such rapid expansion would solve key cosmological problems (such as the horizon and flatness problems, to be discussed shortly), which explains, in part, why his work did not generate the enthusiastic response that Guth's received. In 1981, the Japanese physicist Katsuhiko Sato also developed a version of inflationary cosmology, and even earlier (in 1978), Russian physicists Gennady Chibisov and Andrei Linde hit upon the idea of inflation, but they realized that—when studied in detail—it suffered from a key problem (discussed in note 11) and hence did not publish their work.

The mathematically inclined reader should note that it is not difficult to see how accelerated expansion arises. One of Einstein's equations is $d^2a/dt^2/a = -4\pi/3(\rho + 3p)$ where *a*, ρ, and *p* are the scale factor of the universe (its "size"), the energy density, and the pressure density, respectively. Notice that if the righthand side of this equation is positive, the scale factor will grow at an increasing rate: the universe's rate of growth will accelerate with time. For a Higgs field perched on a plateau, its pressure density turns out to equal the negative of its energy density (the same is true for a cosmological constant), and so the righthand side is indeed positive.

10. The physics underlying these quantum jumps is the uncertainty principle, covered in Chapter 4. I will explicitly discuss the application of quantum uncertainty to fields in both Chapter 11 and Chapter 12, but